



Turn Your Phone Into A Contactless Thermometer

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Due to the outbreak of Covid-19 pandemic, it has become important to check the body temperature of people without any contact with them to prevent spreading of the disease. There are many

contactless thermometers available in the market, but here is a simple solution that will add a thermal screening feature to your smartphone. That is, it will turn your phone into a contactless thermometer using an MLX90614 infrared (IR) sensor module.

You can make a smart Bluetooth based contactless thermometer by connecting the sensor with Bluetooth and getting the body temperature

readings on an app. You can get the sensor data over the phone using a USB cable if you don't wish to use Bluetooth.

This project basically has only two units: transmitter (IR thermometer sensor circuit) and receiver (mobile phone).

For making the Bluetooth based contactless IR thermometer, we need the components listed in the table above.

Coding

To write the Arduino code, install MLX90614 library in the Arduino IDE by selecting Adafruit MLX90614 Library from Sketch→Include Library→Manage Libraries option, and then add it in the code, as shown in Fig. 1. Next, create a setup function and set the baud rate for the Bluetooth module. Here, the default baud rate of Bluetooth is 9600. Create a loop function that runs again and again. Check

Components Required

Component	Quantity	Description
Arduino Micro/Nano/Mini	1	For programming
MLX90614	1	IR temperature sensor
Bluetooth HC-05	1	Bluetooth module
Wires	—	For connections
Battery	1	9V PP3

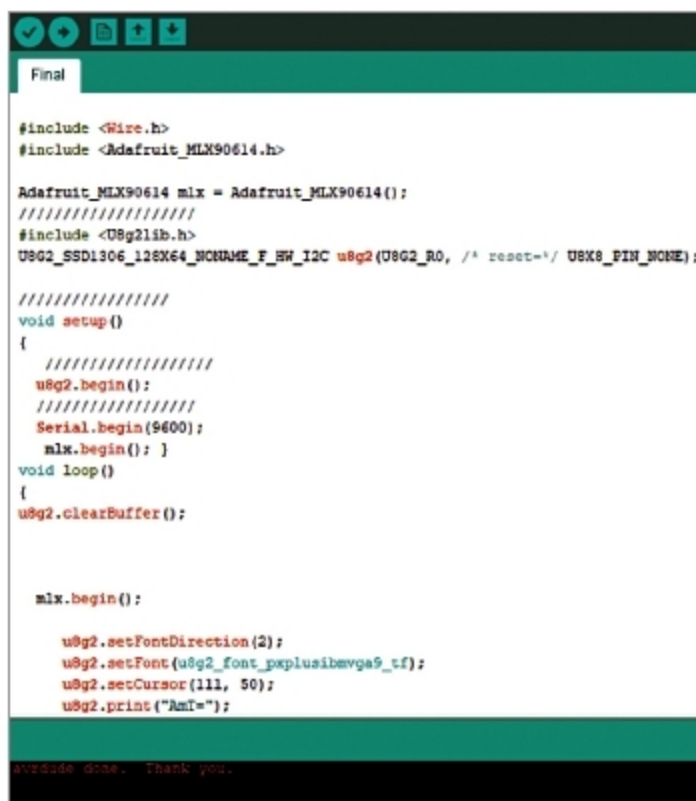


Fig. 1: Arduino code with sensor library

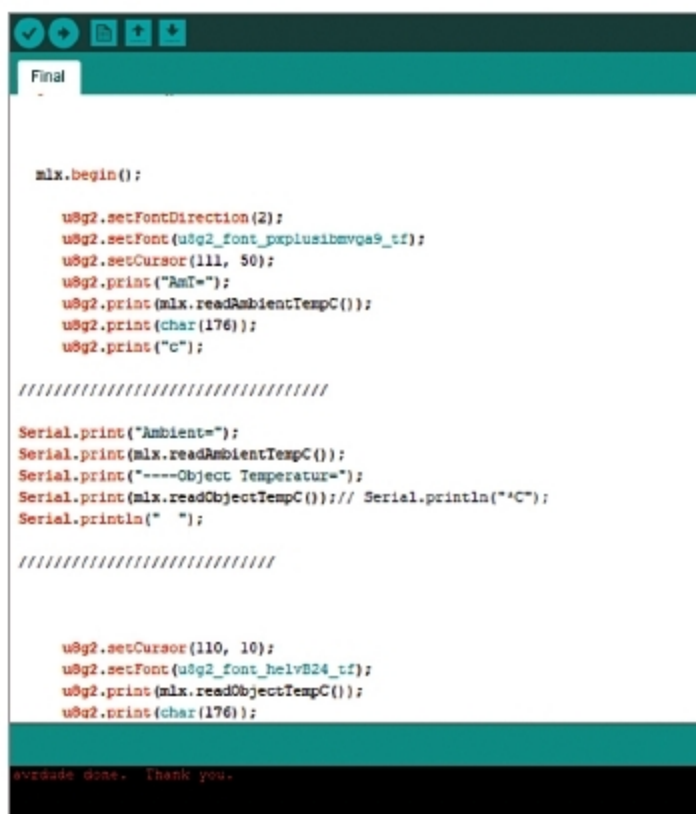


Fig. 2: Serial.println() function

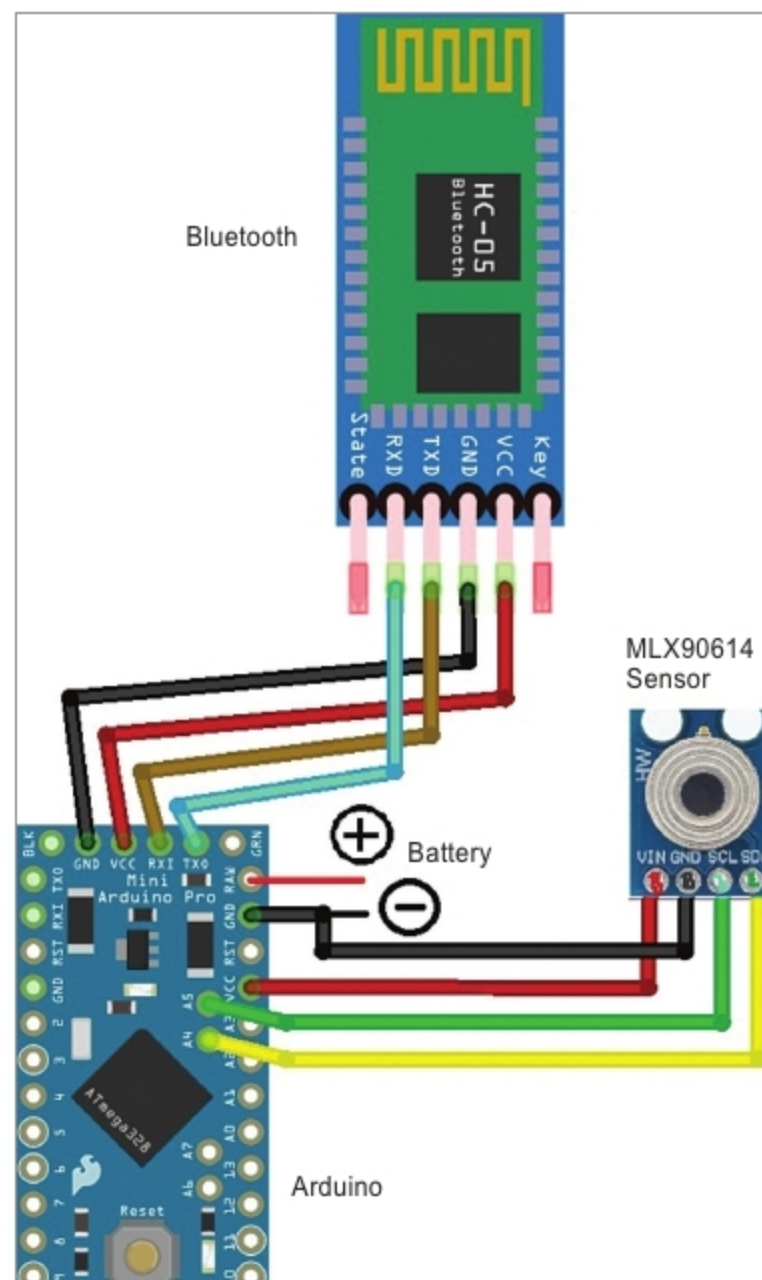


Fig. 3: Connections diagram (transmitter)